1 Intrusion Detection.

* Intrusion is the act of gaining unauthorized access to a system so as to cause loss.

* Intrusion detection is the act of detecting unwanted traffic on a network or a device.

* Intrusion Detection system (IDS) attempt to identify attacks by comparing collected data to predefined signatures known to be malicious or to a model of legal behavior.

Functions of intrusion detection systems.

1. Monitoring and analysis of user and system activity.
2. Auditing of system configurations and vulnerabilities.
3. Assessing the integrity of critical system and data files.
4. Recognition of activity patterns reflecting known attacks.
5. Statistical analysis for abnormal activity patterns.

Process Model.

1. Information sources The different sources of event information used to determine whether an intrusion has takes place.

2. Analysis: The part of intrusion detection system that actually organizes and makes sense of the events derived from the information source, deciding place.

3. Response: The set of actions that the system takes once it detects intrusions. These are typically grouped into active and passive measures.

Benefits of intrusion detection

* Improving integrity of other parts of the information security Infrastructure.
* Improved system monitoring.
* Recognizing and reporting alterations to data file.
* Allowing non-expert staff to contribute to system security
* Providing guidelines on establishing information security Policies.

Types of Intrusion Detection system.

Anomaly Detection.

* An anomaly based intrusion detection system is a system for detecting computer intrusions and misuse by monitoring system activity and classifying it as either normal or anomalous.

* It examines ongoing traffic, activity, transactions, and behaviours in order to identify intrusions by anomalies.

* The classification is based on heuristics or rules, rather than patterns or signatures and will detect.

Advantages.

* Anomaly detectors can produce information that can in turn be used to define signature for misuse detectors.

Disadvantages.

* Anomaly detection approache usually produce a large number of false alarms due to the unpredictable behaviours of users and network.

Signature based Detection.

* A signature based IDS will monitor packets on the network and compare them against a database of signatures or attributes from known malicious threats.

* This is similar to the way most antivirus software detects malware

* New attacks are often unrecognizable by Popular IDS. signatures can be masked as well.

* Also called misuse detection.

Advantages.

* signatures are easy to develop.

* understand if you known what network behavior you're trying to identify.

Disadvantages.

* High false Positive rate

* Largely Ineffective at detecting Previously unknown threats.

* signature database must be continually updated and maintained

Host based IDS (HIDS).

* Host based monitors system logs for evidence of malicious or suspicious application activity in real time.

* Host based IDSs operate on information collected from with in an individual computer system.

* Host based IDS normally utilize information source of two types operating system audit trails, and system logs.

* system logs are much less obtuse and much smaller than audit trails, and are furthermore far easier to comprehend.

Advantage

* with their ability to monitor events local to a host, can detect attacks that cannot be seen by network based IDS.

* It can often operate in an environment in which network traffic is encrypted.

Disadvantages.

* Host based IDS are not well suited for detecting network scans or other such surveillance that targets network.

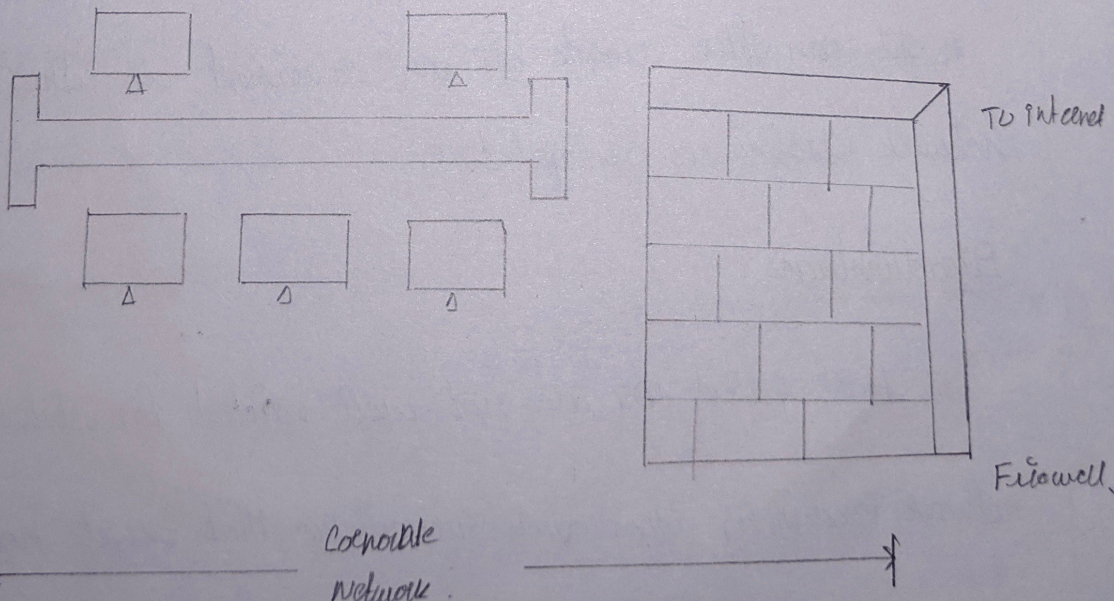
2. Firewalls.

* Informations systems in an organization have changed very rapidly over the years from centralized data processing LANs, WANs and Internet connectivity.

* The Internet connectivity is essential for the organization enabling access to outside world.

* A firewall is inserted between the Internet and LAN for security purpose.

* A firewall may be a hardware or a software program running on a secure host computer. A firewall is placed at junction or gateway between the two networks.



Design goals.

* All the traffic must pass through firewall both from inside to outside and outside to inside.

* only authorized traffic (defined by local security is allowed to pass.

* Firewall itself is immune to penetration.

Service Control: Service Control determines the types of Internet services that are allowed to access both inbound and outbound traffic.

Direction Control: Direction Control determines the direction in which particular service requests may be initiated and is allowed to flow through the firewall.

User control: User control gives access to a service according to which user attempting to access it.

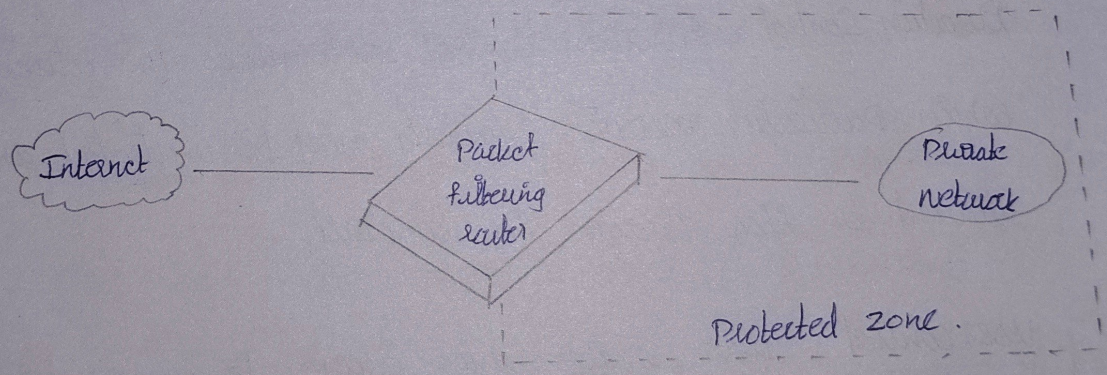
Behavior Control: Behavior control allows to control the use of any particular service.

Types of Firewall.

* Packet filtering router
* Application level gateways.
* Circuit level gateways.

Packet Filtering Router.

* Packet filtering firewalls work at the network level of the osi model, or the layer of TCP/IP.
* This type of firewall only works at the network layer however and does not support rule based models.
* They are usually part of a router.



Advantages.

* Low impact on network performance.
* Packet filters are normally transparent to user.
* Relatively inexpensive price.

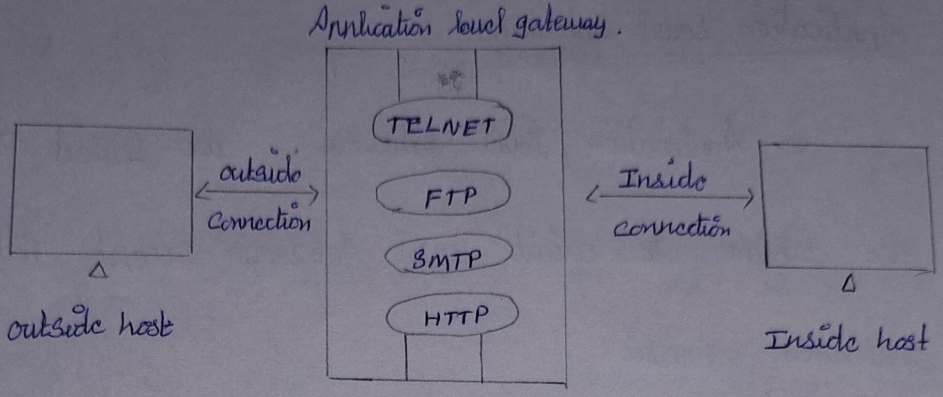
Application Level Gateways.

* Application level gateways, also called Proxies, are similar to circuit level gateways except that they are application specific.

* Incoming or outgoing Packets cannot access services for which is no Proxy.

* In Plain terms, an application layer, they can configured to be a web Proxy will not allow any FTP, gopher, Telnet or other traffic through.

* Because they examine Packets at application layer, they can filter application specific commands such as HTTP: Post and get etc. This cannot be accomplished with either Packet filtering firewalls or circuit level neither of which known anything about the application level information.



Advantages.

* Application gateway provides high level of security than packet filters.
* Easy to Configure.
* They can hide the private network topology.
* It support user level authentication.

Disadvantages.

* High impact on network performance.
* slower in operation because of processing overheads.
* Not transparent to users.

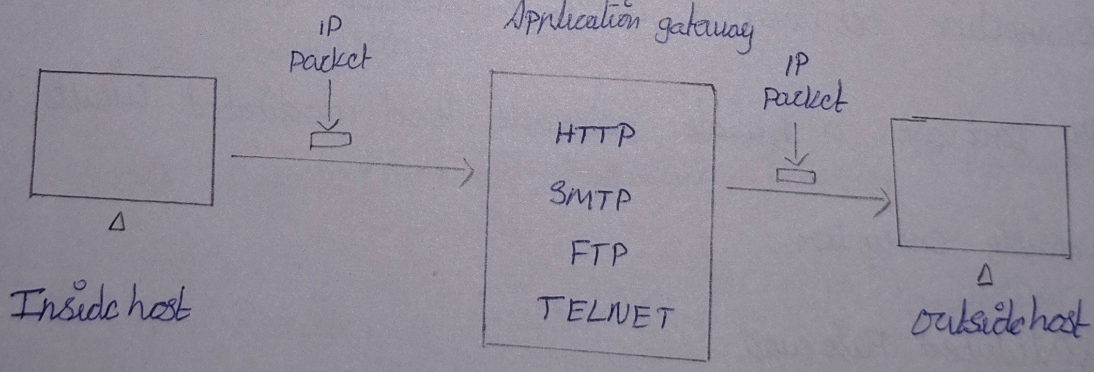
Circuit level Gateways.

* Circuit level gateways work at the session layer of the OSI model, or the TCP layer of TCP/IP.

* They monitor TCP handshaking between Packets to determine whether a requested session is legitimate.

* This is useful for hiding information about Protected networks.

* The circuit level gateway does not permit end-to-end TCP connection but two TCP connection are setup.



Circuit gateway.

Firewall Location

* DMZ network (Demilitarized zone)
* Virtual Private Network.
* Distributed firewall.

DMZ network.

* This allows the DMZ's host to provide service to both the internal and external network while protecting the internal network in case intruders compromise a host in the DMZ.

Virtual Private network.

* Virtual Private Network (VPN) provide an encrypted connection between a user's distributed sites over a public By contrast, a Private network used dedicated circuits and possibly encryption.

Distributed Firewall.

* A distributed firewall configuration involves stand alone firewall devices plus host based firewalls working together under a central administrative.

3. Block Chain.

* Blockchain technology is a decentralized, distributed ledger that stores the record of ownership of digital assests.

* Bitcoin stores all its transaction onto a public database called as blockchain.

* A block chain can be broken down into important components: the block and the chain.

* Attractive properties of Blockchain.

⇒ Log of data with digital signature

⇒ Immutable.

⇒ Cryptographically secure, Privacy Preserving.

⇒ Provides a basis for trusted computing on top of which applications can be built.

Blockchain Technology Layers.

* Blockchain technology consists of five layer: Application and presentation layers, data layer, network layer, consensus layer, infrastructure or hardware layer.

* However, blockchain technology layers can be Layer 0, Layer 1, Layer 3 and Layer 4.

Application and presentation layer.
Smart Contract · Chaincode · DApps · UI

Consensus layer.
Pow · POS · DPOS · POET · PBFT

Network layer.
Peer - to - Peer (P2P).

Data layer
Digital signature · Hash · Merkel tree · Transaction.

Hardware / Infrastruc layer
Virtual machine · Containers · Services · Messaging ·

1. The Hardware Infrastructure layer.

* Blockchain data lies securely stored in a data server. The framework that facilitates this data exchange is known as the client server architecture.

* Blockchain are peer to peer network that allow clients to connect with 'Peer-clients' to make data sharing faster and easier.

2. The Data layer.

* A blockchain is a series of hashed blocks carrying transactional records.

* As thus, in this way, the blockchain keeps on expanding.

3. The Network Layer.

* The P2P framework enables various nodes to exchange transaction data to arrived at a consensus about the validity of a transaction.

* It is the network layer that facilitis this inter node communication. It is also referred to as the 'Propagation layer.'

4) The Consensus Layer.

* This layer is responsible for transaction authentication without this layer, the transaction validation will not take Place.

* Multiple blocks may be formed concurrently, resulting in a branch in the blockchain due to a large number of nodes processing transactions, bundling them and adding them to the blockchain.

5) Application and presentation layer.

* The application layer consists of the programs that end-users take advantage of to establish blockchain network communication. Smart contracts, decentralized application, scripts, user Interface, APIs and Frameworks. Constitus the application layer.

Types of Blockchain Platforms.

* Four different kinds of blockchain architecture are Public, Private, Consortium and hybrid blockchains.

Public Blockchain.

* A public blockchain is a fully decentralized Platform where anyone can read and send transactions. The valid transaction are included in the ledger.

* A public blockchain is a non-restrictive, Permission less distributed ledger system.

Private Blockchain.

* In a Private blockchain, write permission are kept centralized to one organization. In this system the access and permission are tightly controlled and right to modify are restricted to the central authority.

* Private blockchain are more vulnerable to fraudulent activity and malicious actors.

Hybrid Blockchain.

* Hybrid blockchain use both private and public blockchains, rather than being a standalone solution.

* Hybrid blockchain are blockchains that are controlled by a single organization but also have some supervision given by the public blockchain.

Consortium Blockchain.

* Consortium blockchain are permissioned blockchains governed by a group of organization, rather than one entity, as in the case of the private blockchain.

* Consortium blockchains, therefore, enjoy more de decentralization than private blockchains, resulting in the higher levels of security.

4. Cloud security and IOT security.

Cloud security:

* The NIST defines Cloud Computing as a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configureable computing resource that can be rapidly provisioned and released with minimal management efforts or service Provider.

* Cloud provider is responsible for the physical infrastructure and the cloud consumer is responsible for application configuration, Personalization and data.

a) Physical Network → Routers, electrical power cabling, Climate controls, etc.

b) Data storage → Hard drives, etc.

c) Data services → Core network computing hardware & software.

d) Computer virtualization → virtual machine software, host framework. machine and guest machines.

e) Operating system (OS) software that houses.

f) middle ware. Application programming Interface management.

g) Runtime environment Execution and upkeep of a running program.

h) Data All the information stored, modified and accessed.

i) Application Traditional software service.

j) End-user hardware. Computers, mobiles devices, Internet of Thing (IoT) Devices, etc.

Cloud security Challenges and Risk.

1. Data Protection Securing data both at rest and in transit.

2. User Authentication: Limiting access to data and monitoring who accesses the data.

3. Disater and Data Breach: Contingency Planning.

4. Data protection. Data needs to be encrypted at all times, with clearly defined roles when it comes to who will be managing the encryption keys.

5. User Authentication Data resting in the cloud needs to be accessible only by those authorized.

Crucial issues securing the cloud.

The common security issues around cloud computing divided into four main categories.

a) Cloud Infrastructure, Platform and hosted code.

This comprises concerns related to possible virtualization, storage and networking vulnerabilities

b) Data

This category comprises the concerns around data integrity, data lock in, data permanence, provenance and data confidentiality and user privacy specific concerns.

c) Access

This comprises the concern around cloud access, encrypted data communication and user identity management.

d) Compliance

Because of its size and disruptive influence, the cloud is attracting attention from regulatory agencies. especially around security.

IOT Security

* Internet of Things (IOT) security is an approach to safeguard IOT devices connected across a network with protective measures while also preventing cyber attacks.

* IOT security provides blanket protection to networks, system, devices and data from a host of IOT security attacks.

The two key goals of IOT security

1. Make sure all data is collected, stored, Processed and transferred securely.
2. Detect and eliminate Vulnerabilities in IOT components.

* IOT security is part of the organization's overall cybersecurity strategy. It is important to treat connected devices with the same level of security as they would a traditional endpoint, such as a computer or smartphone.

IOT Security Challenges.

* IOT devices encounter several security challenges that pose a risk for organizations and enterprise using them.

Certain notable IOT security challenges:

* Many IOT devices lack built in security. Improper handling of device related security risks, which primarily emerges because these devices don't get regular updates.

* weak credentials and default passwords make devices vulnerable to brute force attacks or password hacking.

* ongoing hybridization of both ransomware and malware strains makes devices vulnerable to different of types of attacks

use of IOT botnets for mining cryptocurrency risks the confidentiality, integrity and availability of data in IOT.